1. Foundations

- $\mathbb{N} = \{1, 2, 3, \ldots\}$
- add, mul on $\mathbb{N}$ (Peano Axioms)
  - commutative
  - associative
  - distributive
- ordering on $\mathbb{N}$

  either $n < m$ or $m < n$

  or $n = m$
- mathmatical induction
- non-empty subset on $\mathbb{N}$

  has least number

- $\mathbb{Z} = \{0, \pm 1, \pm 2, \ldots\}$
- $\mathbb{Q}$: set of rationals

  i.e p/q $(p \in \mathbb{Z}, q \in \mathbb{N})$
- $\mathbb{R}, \mathbb{C}$

---

2. Division Algorithm

- $b|a$ means "$\underbrace{b}_{\text{divisor}}$ divides $\underbrace{a}_{\text{multiple}}$" i.e $a = bc$ $(\exists c \in \mathbb{N})$

- (reflexive) $a|a$
- (transitive) $a|b$, $b|c$ then $a|c$
- (not symmetric) $a|b$, $b|a \iff a=b$
- $b|a$ then $b \le a$

  i.e 자연수는 유한개의 divisors를 가짐.

- divisibility in $\mathbb{Z}$
- "$b \neq 0$ divides a" means $a = bc$ $(\exists c \in \mathbb{Z})$

  $(a, b \in \mathbb{Z})$

$a, b \in \mathbb{Z}$ then $a = bq + r$

~~$(b > 0)$~~

$(b \neq 0)$

where ① $q, r \in \mathbb{Z}$

② $0 \le r < \cancel{b}$ $|b|$

$\because$ $bq \le a < b(q+1)$

$\downarrow \qquad \downarrow$ subtract (bq)

$r < b$

3. Greatest Common Divisor

- $d = (a, b)$ means ① $d \mid a$, $d \mid b$

  ② every common divisor of a, b also divides d

- d is unique $\because$ otherwise $d \mid d'$, $d' \mid d$

- set of natural numbers in form of $(ax+by)$, with $x, y \in \mathbb{Z}$

① $a, b \in (ax+by)$

② let $d = ax + by$ (least number)

③ $a = dq + r$

then $r = a(1-qx) + b(-qy)$

④ $r = 0$, $d \mid a$ $\because$ minimal property of d.

⑤ likewise $d \mid b$

⑥ a, b의 모든 공약수는

$\forall (ax+by) \in \mathbb{N}$을 나눌 수 있어야 한다.

$d = (a, b) \iff$ d is a least number of $\forall (ax+by) \in \mathbb{N}$

- $ax + by = n \iff (a, b) \mid n$

- $(a, b) = 1$ then $ax + by = \forall n$ is solvable (relatively prime)

① $\forall\, a_1, \dots, a_n \in \mathbb{N}$ has

GCD

$$d = (a_1, \dots, a_n)$$
$$= a_1 x_1 + \cdots + a_n x_n$$

② $d = 1$ then ① $a_1, \dots, a_n$ relatively prime

② $a_1 x_1 + \cdots + a_n x_n = n$ always solvable.

4. Euclid Algorithm

• $a = bq_1 + r_1$	$r_n \mid a$	$r_1 = ax_1 + by_1$
• $b = r_1 q_2 + r_2$	$r_n \mid b$	$r_2 = ax_2 + by_2$
• $r_1 = r_2 q_3 + r_3$	$r_n \mid r_1$	$r_3 = ax_3 + by_3$
$\ddots$	$\ddots$	$\ddots$
• $r_{n-2} = r_{n-1} q_n + r_n$	$r_n \mid r_{n-2}$	$r_n = ax_n + by_n$.
• $r_{n-1} = \underbrace{r_n}_{\text{GCD}} q_{n+1} + 0$	$r_n \mid r_{n-1}$	

$187 = 35 \cdot 5 + 12$	$1 \mid 187$	$187 \cdot 3 - 35 \cdot 16 = 1$
$35 = 12 \cdot 2 + 11$	$1 \mid 35$	$-35 \cdot 1 + 12 \cdot 3 = 1$
$12 = 11 \cdot 1 + 1$	$1 \mid 12$	$12 \cdot 1 - 11 \cdot 1 = 1$
$11 = 1 \cdot 11 + 0$	$1 \mid 11$	

$\Rightarrow (187, 35) = 1$

---

5. Fundamental Theorem.

- prime number : ① $\neq 1$

  ② divisible only by itself and 1

$\Rightarrow$ 2, 3, 5, 7, 11 ...

- $n \neq 1$ then least divisor p_1 is prime

  $\frac{n}{p_1} \neq 1$ then least divisor p_2 is prime

  $\frac{n}{p_1 p_2} \neq 1$ then least divisor p_3 is prime.

$\Rightarrow$ $n = p_1 p_2 \cdots p_n$ (finite steps)

$\Rightarrow$ $n = p_1^{j_1} \cdots p_n^{j_n}$ (groupings)

$\Rightarrow$ "unique" (Fundamental Theorem)

- $p \mid mn$ then $p \mid m$ or $p \mid n$ (p is prime)

$\because$ $p \nmid m$ then $(p, m) = 1$ $px + my = 1$

$pnx + mny = n$ therefore $p \mid n$

divisible by p

- $p \mid n_1 n_2 \cdots n_k$ then $p \mid n_l$ $(1 \le l \le k)$

- let $n = p_1^{j_1} \cdots p_k^{j_k}$

- 소수 p'는 모든 decompositional 중.

- $p' = p_l$ $(1 \le l \le k)$ $\because$ $p' \mid p_1^{j_1} \cdots p_k^{j_k}$.

n/p' unique then n unique $\Rightarrow$ induction.

$a = p_1^{\alpha_1} \cdots p_k^{\alpha_k}$ $\Rightarrow$ $(a, b) = p_1^{\gamma_1} \cdots p_k^{\gamma_k}$ where $\gamma_l = \min(\alpha_l, \beta_l)$

$b = p_1^{\beta_1} \cdots p_k^{\beta_k}$ $[a, b] = p_1^{\delta_1} \cdots p_k^{\delta_k}$ where $\delta_l = \max(\alpha_l, \beta_l)$

$\Rightarrow$ $ab = (ab)[a, b]$

6. Properties of Primes

- infinitely many primes $\because$ $p_1 \cdots p_n + 1$

  divisible by prime $\neq p_1, \ldots, p_n$

- $p_n < 2^{2^n}$ $\because$ ① $p_m \mid p_1 \cdots p_n + 1$ $(n < m)$

  ② $p_{n+1} \leq p_1 \cdots p_n + 1$

  $< 2 \cdot p_1 \cdots p_n$

  $< 2 \cdot 2^{(2^1 + \cdots + 2^n)}$

  $= 2 \cdot 2^{2^{n+1} - 2} \leq 2^{2^{n+1}}$

- Prime Number Thm

  Note. $f \sim g$ means $f/g \to 1$

  (f asymptotic to g)

  - $p_n \sim n \log n$
  - $\pi(x) \sim x / \log x$

    : # of primes $\leq x$

- $\sum 1/p_n$ diverges (Euler) i.e infinitely many primes
- $\sum_{p<x} 1/p = \log\log x + c + O(1/\log x)$

- Fermat claimed every $2^{2^n} + 1$ primes $(n = 1, 2, 3 \ldots)$
- False $\because$ $641 \mid 2^{32} + 1$
- construction of regular polygon with p sides (p is prime) $\Longleftrightarrow$ p is Fermat Prime.
- is # of Fermat Primes finite?

- Mersenne Prime : $2^n - 1$ that are primes

① n is prime $\quad \because (2^m)^d - 1 = (2^m - 1)((2^m)^{d-1} + \cdots + 1)$

② useful in searching large primes

e.g $2^{44497} - 1$ is 27th Mersenne prime

with 13,395 digits

- polynomial $f(n)$ with integer coefficients

cannot be prime for every sufficiently large $n \in \mathbb{N}$

(unless f is constant)

- because $f(n)$ divides $f(mf(n)+n)$

Note.

polynomial $\in C^\omega$

테일러 전개 가능

$\because f(\underbrace{mp}_{f(n)} + n) = f(n) + f'(n)(mp) + \frac{f''(n)}{2!}(mp)^2 + \cdots$

- $n^2 - n + 41$ is prime for $n = 1, 2, \ldots, 40$

- there exists polynomial $f(n_1, \ldots, n_k)$ $(n_1, \ldots, n_k \in \mathbb{N})$

s.t primes = set of positive values of f

- related to Hilbert's 10th problem.

- Bertrand's Postulate : $n < \exists \text{prime} < 2n \quad (n>1)$

- $p_{n+1} - p_n = O(p_n^k)$

  with k just a little greater than $1/2$

- $p_{n+1} - p_n$ is not bounded

  $\because$ $n! + m \quad (2 \le m \le n)$ are composite

- Dirichlet Thm

  $(a, \delta) = 1$ then $a, a+\delta, a+2\delta \ldots$

  contains inf many primes

$4n+3$ contains inf many primes

$\because$
- primes $p_1, \ldots, p_n \equiv 3 \pmod 4$
- 考察 $4(p_1 \cdots p_n) - 1 \equiv 3 \pmod 4$
- $\underbrace{1 \cdot 1 \cdot 1 \cdot \cdots \cdot 1}_{\text{prime } 4n+1} \equiv 1$ therefore $p_j \mid 4(p_1 \cdots p_n) - 1$

Two Unsolved Problems

① Goldbach Conjecture : every even integer (>2)

is the sum of two primes

② Twin Prime Conjecture : infinitely many pair of primes

that differ by 2

e.g $(3,5), (17,19) \ldots$

Chen (1974)

" Conjectures are valid

if one of the primes are replaced with

by a number with at most 2-prime factors "

Note. in Goldbach case, the even integer

should be sufficiently large

Eratosthenes : remove multiples of $2, 3, \ldots, \sqrt{n}$

$\Rightarrow$ only primes remain

- Hardy - Littlewood Circle Method

- Vinogradov Thm

: every sufficiently large

odd integer is the sum of three primes